System Modeling And Agile Development Of An Online Bus Reservation System

Course Name: Introduction to Software Engineering

Course Code: IS1103-1

Assignment Number & Title: Assignment 2 – System Modeling and Agile Development of an Online Bus Reservation System

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2. Abstract

This project brings a real-world dimension into designing an online bus reservation system. But the main aim was to understand not only how booking works, but also a distillation of what goes into planning a system ahead of development. It emphasized system modeling and structured development to a greater extent than directly coding.

I developed various UML models to visualize the system. A context diagram was proposed to represent how the system interacts with other parties like users or payment services. Understanding step by step booking process through sequence diagram.

I represented different states (searching, selecting booking & confirmation) using a state diagram.

I made different UML models to show how the system works. I used a context diagram to show how the system works with outside things like users, payment gateways, and bus operators. I made a sequence diagram to show how to book a ticket step by step. A state diagram showed and helps us to understand the different stages of a booking, such as searching, choosing correctly, correct payment without any issues, and final closing.[1]

Used scrum to make agile methodology actually work in our day to day work. I kept things moving by breaking the work into sprints and tailoring user stories for each group. I knew exactly what they needed to do and what the finish line looked like.

This assignment helped me understand how software systems are planned, designed, and built in a logical way. Overall, this assignment gave a basic idea of how software systems are planned, analysed, and improved step by step instead of directly jumping into coding [2].

3.Github Repository Link

GitHub Url : https://github.com/ Bhavyatha21/ISE-Task-2: System Modeling And Agile Development Of An Online Bus Reservation System

4. Introduction

In software engineering, designing a system before implementation is very important. Many times, development starts without proper structure, and later it becomes difficult to manage. This assignment was done to understand how system modelling helps in avoiding such problems.

The main idea was to design an online bus reservation system and represent it using UML diagrams. These diagrams are useful because they give a clear picture of the system without writing actual code [1]. It becomes easier to understand how users interact with the system and how data flows.

Agile development was also included in this assignment. Scrum was used to divide the work into smaller parts called sprints. This approach is commonly used in real projects because it allows continuous improvement [2].

This assignment is related to system modelling and Agile methods unit. It helped in understanding how theory is applied in practical situations.

5. Problem Statement

The goal is to make buying tickets online quick, user-friendly and to build a seamless booking experience that lets people to secure their bus seats in just few clicks and it also saves their time and stress.

The system should let :

- Looking for buses between places
- Choosing seats
- Paying
- Looking at booking information
- Cancelling tickets

The goal is to have a clear system design and a good Agile development plan.

Some things were taken for granted:

- People can get online

- The payment gateway works as it should.
- Users data is correct

6. Dataset Description

This assignment is not relevant because it is about designing systems, not analysing data. So dataset-related information details are not applicable here.

7. Tools and technologies that were used

Programming Language: Not needed

Tools Used :

Draw.io (for making UML diagrams)

GitHub (for keeping track of changes)

Microsoft Word (for writing documents)

8. Methodology / Architecture of the Model

8.1 Description of the Model

Different models were used to understand the system in a better way. Each model gives a different perspective.

The context model helps us to understand how the system interacts with external entities. It gives a basic overview and its applications without going into too much detail. This helps in identifying system boundaries [1].

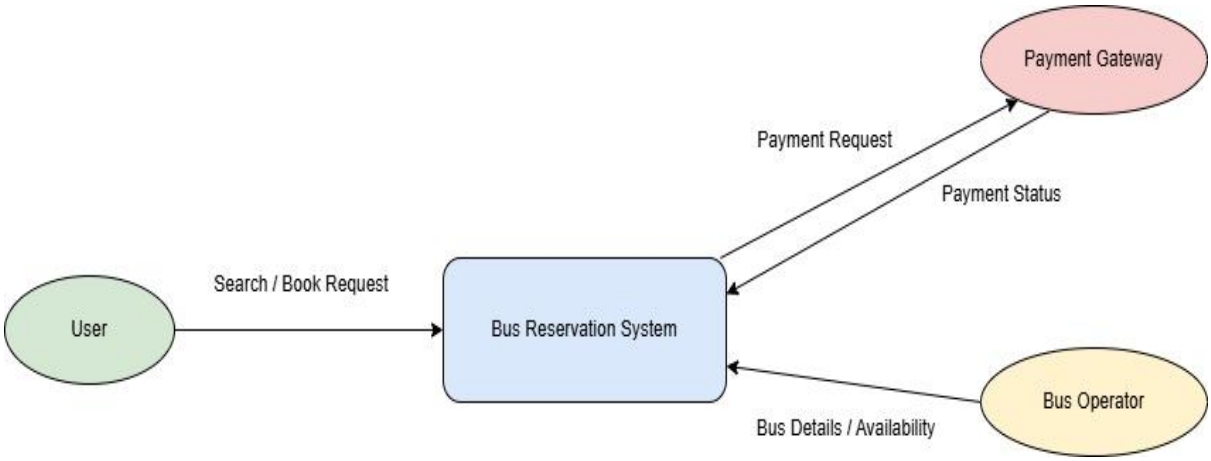
The sequence diagram shows the order of operations. It explains how the user interacts with the system step by step. It is useful to understand the flow of actions during booking [3].

The state diagram shows different states of a booking. For example, searching, selecting seats, payment, and booking. It also shows how the system moves from one state to another [1].

Altogether, these diagrams make the system and the users easy to understand even without looking at code.

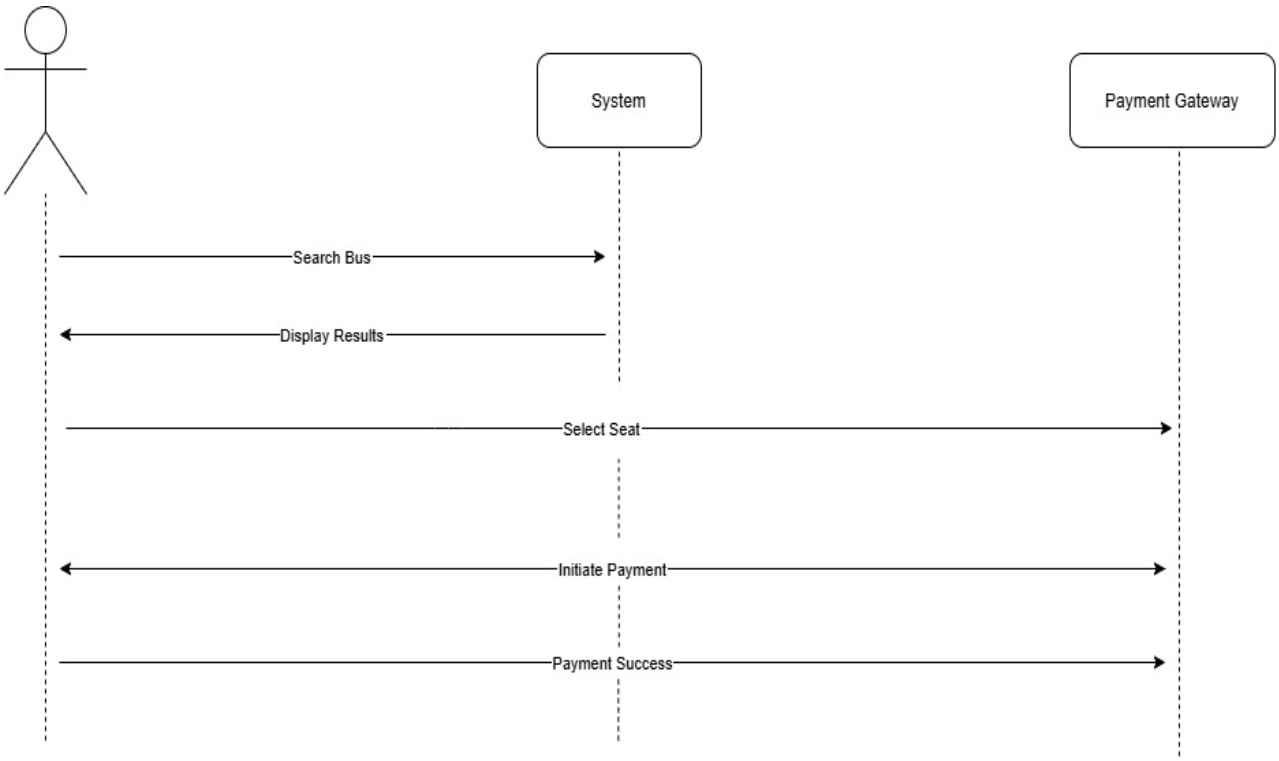
8.2 Diagrams

a) Context Diagram



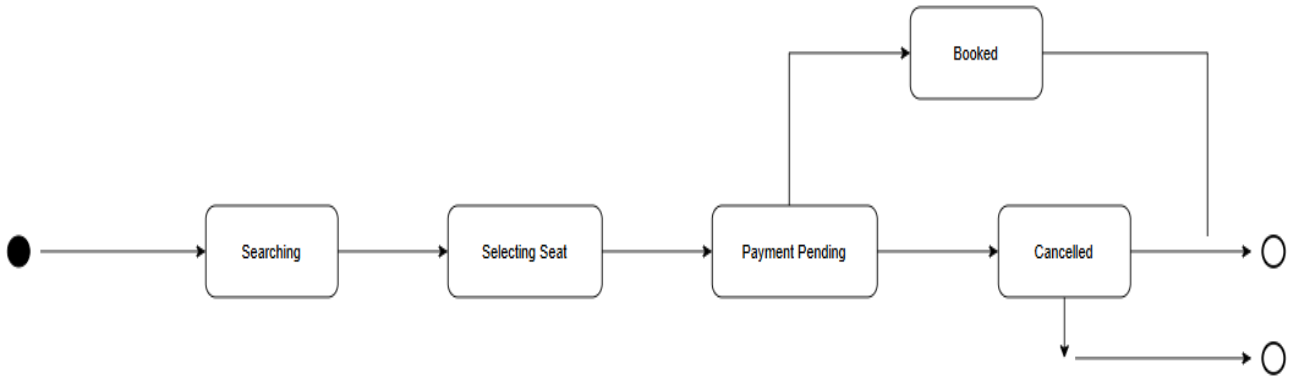
This diagram helps us to understand how the user and the system work together and analyses how the system connects to other different services. It gives you a basic idea of where the system ends.

b) Diagram of the Sequence (Booking Process)



This diagram shows how things happen when you book a ticket. It made it easy to see each step, from searching to confirming.

c) Diagram of the State (Booking lifecycle)



This diagram shows the different stages of a booking. It shows how the system goes from one step to the next.

9. Implementation Details

This assignment was mostly about design, so there was no actual coding. But the system was broken up into logical parts.

The main parts are:

- Module for users (login and search)
- Module for booking
- Module for payments
- Module for admins

If these modules are put into use, they can be made separately and then put together later.

If implemented, the system can follow a layered architecture with frontend, backend, and database. This type of structure is commonly used in real applications [2].

10. Experiments Performed

Because this isn't a machine learning task, the experiments were more like ways to make the design better.

Different methods were tried to make it easier to book. We changed the order of the steps to make it clearer. To keep the system simple, some steps that weren't needed were taken out.

11. Results and Analysis

11.1 Quantitative Results

This assignment doesn't have any numbers because it doesn't involve data analysis or machine learning. The assessment hinges on the precision and lucidity of system models.

11.2 Graphs and Plots

Graphs and plots are not applicable. UML diagrams were used instead to represent the system.

11.3 Qualitative Results

Using the system modelling approach made it easier to understand the application in a clear way.

- The context diagram makes it easy to see how systems work together.
- The sequence diagram shows how to book step by step.
- The state diagram shows all the different states that a booking can be in.

The design of the system is user-friendly, simple, clear, and easy to understand and follow .

12. Discussion

Designing a system takes a lot of thought and it became clear and I realised it while I was working on this assignment. There are many steps and interactions in even a simple app.

At first, they didn't think about steps like confirming payment. These gaps were only visible after I drew diagrams. Agile methodology also helped me figure out how to break big tasks down into smaller ones.

While doing this assignment, it was noticed that even a simple system has many internal steps. At first, some parts were missed, like payment confirmation. But after drawing diagrams, these issues became visible.

Agile methodology also helped in organizing the work. Instead of doing everything at once, tasks were divided into smaller parts. This made the process easier to manage [2].

13. Regularization and Optimization

Not relevant for this task.

14. Limitations and Future Work

The system that was made is simple and doesn't have any advanced features like:

- Updates on seat availability in real time
- Pricing that changes
- Systems for making suggestions

These features can be added in the future to make the system better. In future, these features can be added to improve the system. More security features can also be included.

15. Conclusion

This task helped me learn about system modelling and Agile development methods. It showed how important it is to plan before you do something.

UML diagrams helped me see how the system worked. Using the Scrum method helped organize the development process well.

In general, the assignment helped me understand software engineering ideas better.

Overall, this assignment gave a clear idea of how real systems are designed and improved over time [1].

16. References

[1] I. Sommerville, *Software Engineering*, 10th Edition, Pearson, 2016.
[2] Agile Alliance, "Agile Methodology Guide."
[3] Draw.io Documentation

17. Appendix

Additional diagrams, draft designs, and supporting and necessary notes related to the system modelling can be included here if required. In this assignment, I have added all the necessary details and all important information has been covered in the main sections.

18. Declaration of Academic Integrity

I hereby declare that this assignment is my original work and has not been copied or plagiarized from any source. All references and resources used have been properly cited. Any ambiguities will lead to forfeiting my marks.

Student Signature

Date: April 03, 2026